

Comparative Assessment of Automated and Manual Monitoring in Comprehensive Plant–Pollinator Communities

immediate

Supplementary Material

Table S1: Pollinator functional groups, included in the YOLOv5 training dataset, indicating the number of individuals, and the representative species in each group.

Functional Group	N° Individuals	Representative Species
Large Bees	1,674	<i>Amegilla quadrifasciata</i> , <i>Apis mellifera</i> , <i>Anthophora balearica</i> , <i>Bombus terrestris</i> , <i>Bombus hispanica</i> , <i>Colletes</i> sp.
Small Bees	1,223	<i>Andrena</i> sp., <i>Hyaleus</i> sp., <i>Lasioglossum</i> sp., <i>Nomioides</i> sp.
Hoverflies	523	<i>Eristalis</i> sp., <i>Eristalinus</i> sp., <i>Eupeodes</i> sp., <i>Scaeva</i> sp., <i>Ischiodon</i> sp., etc.
Lepidoptera	428	<i>Autographa gamma</i> , <i>Lycaena</i> sp., <i>Macroglossum stellatarum</i> , <i>Vanessa</i> sp.
Dipterans	201	<i>Exoprosopa</i> sp., <i>Musca</i> sp., <i>Pyrrellia</i> sp., <i>Sarcophaga</i> sp.
Wasps	201	<i>Ancistrocerus</i> sp., <i>Megascolia</i> sp., <i>Polistes</i> sp., etc.
Coleopterans	189	<i>Anaspis</i> sp., <i>Attagenus trifasciatus</i> , <i>Dasytes</i> sp., etc.
Small Dipterans	216	<i>Pthirrita</i> sp., <i>Acanthiophilus helianthi</i> , <i>Agromyzidae</i> sp., <i>Sphenella marginata</i>
Small Coleopterans	216	<i>Meligethes</i> sp., <i>Mordellistena</i> spp., <i>Psylliodes</i> sp., etc.
Other Arthropods	61	Mostly ants, Thysanoptera, Hemiptera, etc.
Unknown insect	384	No functional group assigned
Moths	17	No species assigned

Table S2: Plant species included in the YOLOv5 training dataset, indicating the number of individuals, data source, island of collection, and year of record. The ACS refers to the Automatic Camera System, with three configurations: V2 (wide-angle lens), HQ (high-quality lens with 35mm focal length), and NoIR (wide nocturnal lens equipped with infrared sensors).

Plant sp	N° individuals	Source	Island	Year
<i>Cakile maritima</i>	3	ACS-V2	Menorca	2021
<i>Chrysanthemum coronarium</i>	2	ACS-V2	Menorca	2021
<i>Cistus monspeliensis</i>	2	ACS-V2	Menorca	2021
<i>Convolvulus</i> sp.	2	ACS-V2	Menorca	2021
<i>Dorycnium fulgurans</i>	2	ACS-V2	Menorca	2021
<i>Euphorbia serrata</i>	2	ACS-V2	Menorca	2021
<i>Helianthemum</i> sp.	2	ACS-V2	Menorca	2021
<i>Launea cervicornis</i>	2	ACS-V2	Menorca	2021
<i>Lavatera maritima</i>	3	ACS-V2	Cabrera	2021
<i>Reichardia picroides</i>	2	ACS-V2	Menorca	2021
<i>Santolina chamaecyparissus</i>	2	ACS-V2	Menorca	2021
<i>Teucrium capitatum</i>	4	ACS-V2	Menorca	2021

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Table S2: (Continued)

Plant sp	N° individuals	Source	Island	Year
<i>Allium ampeloprasum</i>	1	ACS-V2	Cabrera	2022
<i>Allium</i> sp.	1	ACS-V2	Cabrera	2022
<i>Anacamptis pyramidalis</i>	1	ACS-V2	Cabrera	2022
<i>Asteriscus aquaticus</i>	1	ACS-V2	Cabrera	2022
<i>Astragalus balearicus</i>	1	ACS-V2	Cabrera	2022
<i>Blackstonia perfoliata</i>	1	ACS-V2	Cabrera	2022
<i>Cakile maritima</i>	1	ACS-V2	Cabrera	2022
<i>Cistus monspeliensis</i>	1	ACS-V2	Cabrera	2022
<i>Daucus carota</i>	1	ACS-V2	Cabrera	2022
<i>Dorycnium fulgurans</i>	1	ACS-V2	Cabrera	2022
<i>Euphorbia dendroides</i>	1	ACS-V2	Cabrera	2022
<i>Euphorbia serrata</i>	1	ACS-V2	Cabrera	2022
<i>Fumana ericoides</i>	1	ACS-V2	Cabrera	2022
<i>Gladiolus communis</i>	1	ACS-V2	Cabrera	2022
<i>Globularia alypum</i>	1	ACS-V2	Cabrera	2022
<i>Lavatera arborea</i>	1	ACS-V2	Cabrera	2022
<i>Limonium</i> sp.	1	ACS-V2	Cabrera	2022
<i>Ruta</i> sp.	1	ACS-V2	Cabrera	2022
<i>Sedum sediforme</i>	1	ACS-V2	Cabrera	2022
<i>Teucrium capitatum</i>	1	ACS-V2	Cabrera	2022
<i>Allium ampeloprasum</i>	1	ACS-HQ	Cabrera	2023
<i>Allium</i> sp.	1	ACS-HQ	Cabrera	2023
<i>Asteriscus aquaticus</i>	1	ACS-HQ	Cabrera	2023
<i>Astragalus balearicus</i>	1	ACS-HQ	Cabrera	2023
<i>Blackstonia perfoliata</i>	1	ACS-HQ	Cabrera	2023
<i>Cakile maritima</i>	1	ACS-HQ	Cabrera	2023
<i>Cistus monspeliensis</i>	1	ACS-HQ	Cabrera	2023
<i>Daucus carota</i>	1	ACS-HQ	Cabrera	2023
<i>Dorycnium fulgurans</i>	1	ACS-HQ	Cabrera	2023
<i>Euphorbia dendroides</i>	1	ACS-HQ	Cabrera	2023
<i>Euphorbia serrata</i>	1	ACS-HQ	Cabrera	2023
<i>Fumana ericoides</i>	1	ACS-HQ	Cabrera	2023
<i>Globularia alypum</i>	1	ACS-HQ	Cabrera	2023
<i>Limonium</i> sp.	1	ACS-HQ	Cabrera	2023
<i>Ruta</i> sp.	1	ACS-HQ	Cabrera	2023
<i>Sedum sediforme</i>	1	ACS-HQ	Cabrera	2023
<i>Teucrium capitatum</i>	1	ACS-HQ	Cabrera	2023
<i>Salvia rosmarinus</i>	1	ACS-NoIR	Cabrera	2023
<i>Dittrichia viscosa</i>	1	ACS-NoIR	Cabrera	2023
<i>Urginea maritima</i>	1	ACS-NoIR	Cabrera	2023
<i>Cakile maritima</i>	4	ACS-HQ	Formentera	2024
<i>Daucus carota</i>	2	ACS-HQ	Formentera	2024
<i>Echium simplex</i>	3	ACS-HQ	Formentera	2024
<i>Helichrysum stoechas</i>	2	ACS-HQ	Formentera	2024
<i>Lotus cytisoides</i>	2	ACS-HQ	Formentera	2024
<i>Micromeria</i> sp.	2	ACS-HQ	Formentera	2024
<i>Sonchus tenerrimus</i>	2	ACS-HQ	Formentera	2024
<i>Teucrium capitatum</i>	2	ACS-HQ	Formentera	2024
<i>Thymus</i> sp.	3	ACS-HQ	Formentera	2024
<i>Aloe vera</i>	1	ACS-HQ	Eivissa	2024

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Table S2: (Continued)

Plant sp	N° individuals	Source	Island	Year
<i>Asphodelus fistulosus</i>	1	ACS-HQ	Eivissa	2024
<i>Bellardia trixago</i>	1	ACS-HQ	Eivissa	2024
<i>Bituminaria bituminosa</i>	1	ACS-HQ	Eivissa	2024
<i>Carlina</i> sp.	1	ACS-HQ	Eivissa	2024
<i>Chrysanthemum coronarium</i>	1	ACS-HQ	Eivissa	2024
<i>Cistus albidus</i>	1	ACS-HQ	Eivissa	2024
<i>Convolvulus</i> sp.	1	ACS-HQ	Eivissa	2024
<i>Diploaxis</i> sp.	1	ACS-HQ	Eivissa	2024
<i>Echium simplex</i>	1	ACS-HQ	Eivissa	2024
<i>Erica multiflora</i>	1	ACS-HQ	Eivissa	2024
<i>Foeniculum vulgare</i>	1	ACS-HQ	Eivissa	2024
<i>Galactites tomentosa</i>	1	ACS-HQ	Eivissa	2024
<i>Launea cervicornis</i>	1	ACS-HQ	Eivissa	2024
<i>Lavandula dentata</i>	1	ACS-HQ	Eivissa	2024
<i>Phlomis</i> sp.	1	ACS-HQ	Eivissa	2024
<i>Reichardia picroides</i>	1	ACS-HQ	Eivissa	2024
<i>Reseda</i> sp.	1	ACS-HQ	Eivissa	2024
<i>Rubus ulmifolius</i>	1	ACS-HQ	Eivissa	2024
<i>Salvia rosmarinus</i>	1	ACS-HQ	Eivissa	2024

Table S3: Performance metrics for the leave-one-out experiments. Each row corresponds to a test in which the indicated plant species was included only in the test set—meaning no examples from that species were used during training or validation. P refers to *Precision* and R to *Recall*. The *Validation* columns report results on the validation set, while the *Test* columns show the outcomes obtained during the test stage.

Model	Validation				Test			
	P	R	mAP@50	F1	P	R	mAP@50	F1
<i>Cakile maritima</i>	0.97	0.96	0.98	0.96	0.69	0.43	0.50	0.53
<i>Helichrysum stoechas</i>	0.94	0.92	0.96	0.92	0.70	0.31	0.44	0.43
<i>Lotus cytisoides</i>	0.93	0.91	0.95	0.91	0.58	0.66	0.55	0.61
<i>Sonchus tenerrimus</i>	0.94	0.94	0.97	0.94	0.94	0.90	0.93	0.91
<i>Daucus carota</i> , <i>H. stoechas</i> , <i>Reichardia picroides</i>	0.93	0.91	0.94	0.91	0.88	0.51	0.62	0.64

Table S4: Comparison of pollinator detections by YOLOv5 and manual review across diurnal and nocturnal video samples (Videos 1 to 12). Detection counts are shown as YOLOv5 / Manual Review. The number of false positives for each video is reported in the bottom row. Videos were recorded using the Automatic Camera System, with three configurations: V2 (wide-angle lens), HQ (high-quality lens with 35mm focal length), and Noct (wide nocturnal lens equipped with infrared sensors).

Pollinator	1 (V2)	2 (V2)	3 (HQ)	4 (V2)	5 (V2)	6 (HQ)	7 (HQ)	8 (V2)	9 (V2)	10 (NoIR)	11 (NoIR)	12 (NoIR)
<i>Anthophora</i> sp.	4 / 4		6 / 6	6 / 6	1 / 1	3 / 3	4 / 5		2 / 3			
<i>Colletes</i> sp.			4 / 4			5 / 5		2 / 2				
<i>Nomioides</i> sp.			1 / 1			1 / 1						
<i>Sarcophaga</i> sp.		1 / 1			2 / 2							
Microdiptera sp.			3 / 4	1 / 3		0 / 1	1 / 2	0 / 1				
<i>Eristalinus</i> sp.					2 / 2							
<i>Lasioglossum</i> sp.			1 / 1			1 / 1						
Moth										3 / 3	2 / 2	5 / 5
False positives	2	2	0	1	0	1	0	1	0	0	0	2

Table S5: Results of Wilcoxon signed-rank (Paired Wilcoxon test) tests comparing methods for the Connectance, Network-level Specialisation and Robustness metrics by habitat and year

Group	Metric	N pairs	W statistic	p-value
Habitat				
Dune system	Connectance	17	61	0.74
	Network-level Specialisation	17	83	0.45
	Robustness	11	47	0.24
Pine forest	Connectance	10	7	0.14
	Network-level Specialisation	10	12	0.13
	Robustness	7	24	0.11
Rocky coastal	Connectance	22	51	0.03*
	Network-level Specialisation	22	46	0.01*
	Robustness	17	116	0.04*
Year				
2022	Connectance	23	83	0.27
	Network-level Specialisation	23	143	0.60
	Robustness	17	91	0.517
2023	Connectance	26	72	0.04*
	Network-level Specialisation	26	45	< 0.01*
	Robustness	18	158	< 0.01*

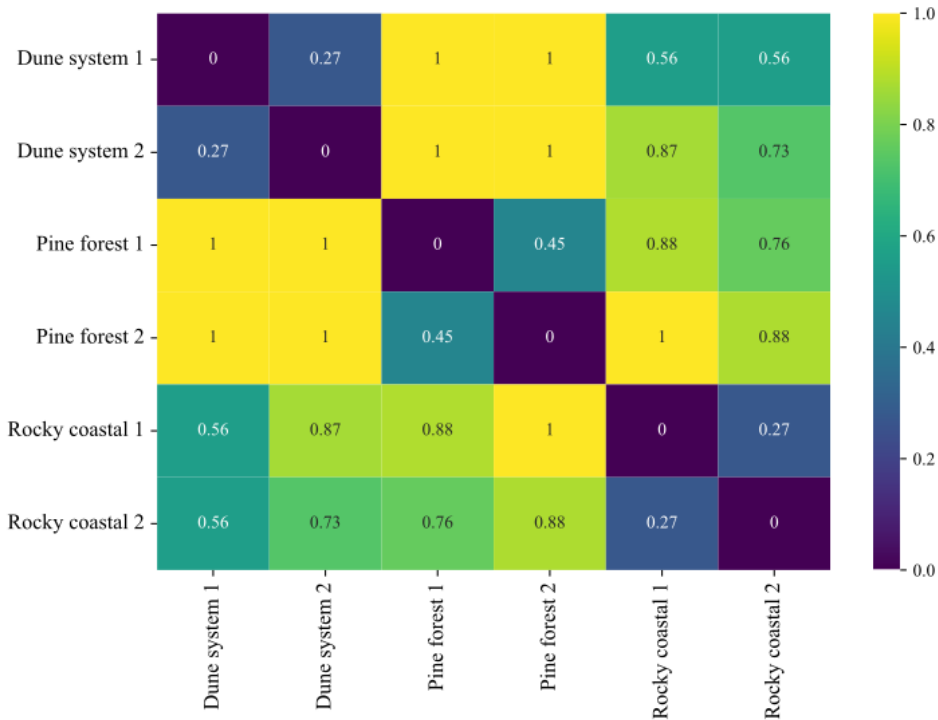


Figure S1: Bray-Curtis dissimilarity heat-map comparing plant-species composition across habitats. Values range from 0 (identical assemblages) to 1 (completely dissimilar), with warmer colours indicating greater compositional divergence. Each cell reports the pairwise Bray-Curtis distance between study sites.

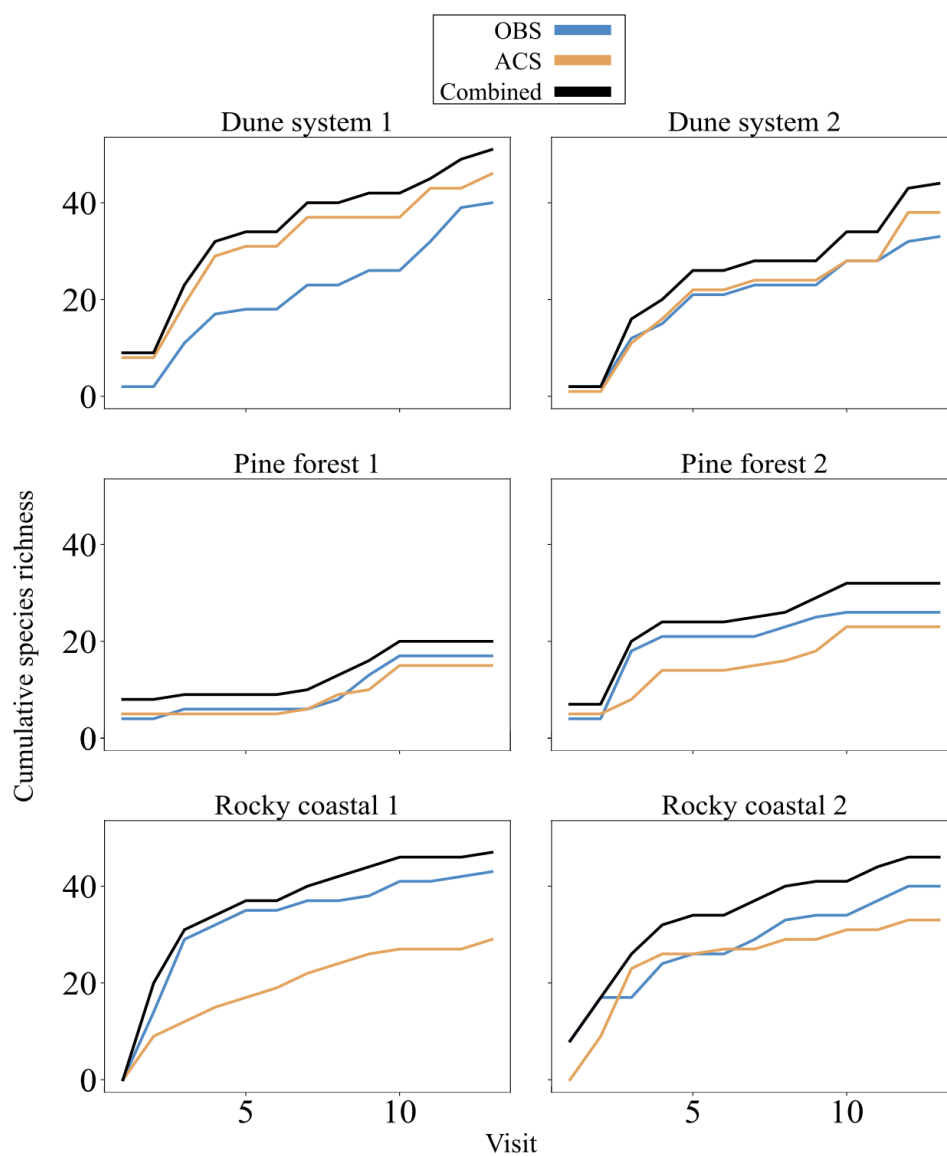


Figure S2: Cumulative pollinator species richness per visit across study sites for each method: direct observations (blue), Automatic Camera System (orange) and both methods merged (black). The curves show how species richness increases with sampling effort, allowing comparison of completeness between focal observations.